

Technical Paper

Structural Significance and Importance of External Verification Architecture (EVA) in Artificial Intelligence

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Abstract

The safety assurance of current AI systems relies heavily on self-assessment conducted by the developers themselves. International regulations and standards — including the EU AI Act and ISO/IEC 42001 — demand independent external verification, yet the concrete architectural framework required to technically realize such verification has not yet been systematically defined.

This paper proposes the concept of External Verification Architecture (EVA) to fill this gap. EVA is a technical framework for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning processes, outputs, and decision-making structures of AI systems through mechanisms that are independent of the model's internal parameters.

EVA is defined by three structural conditions: Verifiability, Transparency, and Physical Immutability. Only when all three conditions are satisfied simultaneously can genuine external verification be achieved — one in which the model cannot falsify, circumvent, or disable its own verification mechanism.

International standardization is indispensable for the social implementation of EVA. Without integration into ISO/IEC JTC 1/SC 42, CEN-CENELEC JTC 21, and the EU AI Act technical specifications, EVA will remain confined to individual implementations and cannot serve as a cross-industry trust infrastructure. This paper examines EVA's standardization pathways and alignment with major standards bodies in detail.

Finally, this paper argues that the Human Natural Structure (HNS) framework constitutes one viable technical method for realizing EVA. While HNS should be evaluated independently of EVA's conceptual definition and standardization discussion, it is positioned as an implementation pathway that can concretely provide the structural reference baseline that EVA requires.

1. Introduction

As AI systems are rapidly deployed across society, there is growing international demand for frameworks that objectively guarantee their safety, reliability, and ethical compliance. Yet in practice, AI safety assurance is predominantly dependent on internal evaluations designed and conducted by the model developers themselves — so-called self-assessment — and no established mechanism exists for independent third-party technical verification.

This represents a clear institutional gap when compared to the independence requirements of clinical trials in the pharmaceutical industry or external audits in accounting. Despite the EU AI Act and ISO/IEC 42001 explicitly requiring 'external verification,' the technical implementation methods remain unspecified. Closing the gap between regulatory requirements and technical implementation is the starting point of this paper.

This paper proposes the concept of External Verification Architecture (EVA) and systematically examines its technical definition, implementation pathways, and significance in international standardization. EVA does not negate existing AI safety, audit, or explainability research; rather, it is positioned as an architecture that fulfills the function of 'continuous structural external verification' that existing approaches are structurally unable to provide.

The paper is organized as follows. Section 2 identifies the limitations of current AI safety assurance. Section 3 presents EVA's conceptual definition and three structural conditions. Section 4 discusses EVA's technical implementation pathways. Section 5 details EVA's

significance in international standardization and concrete pathways. Section 6 briefly discusses the significance of Human Natural Structure (HNS) as one technical method for realizing EVA. Section 7 concludes.

2. Limitations of Current AI Safety Assurance

2.1 The Structural Problem of Self-Assessment Dependency

The mainstream of current AI safety evaluation consists of internal assessments designed and conducted by model developers. Representative methods include Red Teaming, RLHF (Reinforcement Learning from Human Feedback), Constitutional AI, and internal benchmark evaluation — all of which have produced meaningful results in advancing both model capability and safety.

However, these methods share a common structural limitation: the design, execution, and interpretation of evaluation results all fall within the discretion of the same developer, with no institutional guarantee of independent third-party verification. More fundamentally, no matter how sincere the model developer, self-assessment carries a structural tendency to overlook risks that the developer themselves has not anticipated.

2.2 The Absence of Technical Verifiability

A key technical barrier to external verification is the 'opacity' of AI reasoning processes. Modern large language models have billions to hundreds of billions of parameters, constituting a black box from which it is extremely difficult to causally explain externally why a particular output was generated.

For an external verification mechanism to render 'judgments,' structured data constituting the verification target must be generated in real time, in parallel with inference. Current AI architectures are not designed with the assumption of connectivity to such external verification mechanisms. This is not a technical accident but the result of a design choice — and EVA aims to fill this design-level gap.

2.3 The Gap Between Regulatory Requirements and Technical Implementation

The EU AI Act mandates technical documentation, log-keeping, and human oversight for high-risk AI systems. However, the technical specifications for 'log-keeping' are not stipulated: what to record, at what granularity, and in what format are left to the developer's interpretation.

ISO/IEC 42001:2023 provides a framework for AI management systems, but its scope of certification covers processes, not independent technical verification of reasoning processes themselves. Closing this gap between regulatory requirements and technical implementation is EVA's central mission.

3. Conceptual Definition of External Verification Architecture (EVA)

3.1 Definition of EVA

External Verification Architecture (EVA) is a technical framework for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning processes, outputs, and decision-making structures of AI systems through external mechanisms independent of the model's internal parameters.

The essence of EVA lies in the 'structural guarantee of externality.' As long as the verification mechanism depends on the same developer, the same software environment, or the same hardware substrate as the model itself, that verification is effectively no different from self-assessment. True external verification must carry a structural guarantee that the model cannot falsify, circumvent, or disable its own verification mechanism.

3.2 The Three Structural Conditions of EVA

For EVA to function effectively, the following three conditions must be satisfied simultaneously.

Condition	Definition	Technical Implication
Verifiability	Each reasoning step is output externally in machine-readable format	Structured logs in JSON-LD / RDF standard formats. Third parties must be able to analyze programmatically
Transparency	A monitoring mechanism operates in parallel with inference, generating an auditable trail	Operates as a process independent of model inference. Designed so the model cannot recognize or manipulate the monitoring mechanism
Physical Immutability	The verification logic itself is implemented on a substrate that cannot be altered by the model	Embedded in ROM / FPGA / ASIC. Implementation at the hardware layer inaccessible via software updates

These three conditions constitute the logical minimum requirements for EVA. Verifiability alone cannot eliminate the risk of logs being subsequently falsified. Transparency alone leaves the possibility that the model intentionally circumvents the verification mechanism. Physical Immutability alone allows the criteria for what is verified to become arbitrary. Only when all three conditions overlap does genuine external verification become possible.

3.3 Relationship Between EVA and Existing Related Concepts

EVA is related to existing concepts such as Explainable AI (XAI), AI auditing, and model cards, but differs from them in the following essential respects.

Existing Concept	Primary Focus	Essential Difference from EVA
Explainable AI (XAI)	Post-hoc interpretation and visualization	EVA is real-time, pre-designed structural verification
AI Auditing	Periodic, snapshot-based evaluation	EVA provides continuous, structural per-inference recording
Model Cards	Documentation of capabilities and limitations	EVA is a dynamic, runtime verification mechanism

ISO/IEC 42001	Certification of management processes	EVA enables external verification of reasoning processes themselves
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EVA does not replace these existing concepts; rather, it is an architecture that newly fulfills the functional layer of 'continuous external verification at inference time' that existing approaches are structurally unable to cover.

3.4 What EVA Achieves

The fundamental change EVA achieves is a shift in AI safety assurance from assertion to proof. Current AI safety relies on assertions by developers that their models are safe. EVA provides the technical foundation for recording and proving those assertions in a form verifiable by independent third parties.

This is functionally analogous to the flight recorder on an aircraft or the audit log of financial transactions. As AI systems become entrenched as social infrastructure, such verifiable recording mechanisms are indispensable as a foundation of social trust — beyond mere regulatory compliance.

4. Technical Implementation Pathways of EVA

4.1 The Three-Layer Implementation Structure

Full implementation of EVA is achieved by integrating three layers: the software layer, the hardware layer, and the standards layer. Each layer has independent value, but it is the integration of all three that fully satisfies EVA's three conditions of Verifiability, Transparency, and Physical Immutability.

Implementation Layer	EVA Conditions Satisfied	Key Technologies
Software Layer	Verifiability, Transparency	Sidecar architecture, JSON-LD structured logging, W3C PROV-O compliance
Hardware Layer	Physical Immutability	FPGA / ASIC embedding, TPM integration, Secure Enclave (Intel SGX, etc.)
Standards Layer	Social trust, cross-industry adoption	ISO/IEC SC 42, CEN-CENELEC JTC 21, EU AI Act technical specifications

4.2 Software Layer: Sidecar Architecture

EVA's software layer is implemented as a sidecar architecture operating in parallel with model inference. This sidecar writes the following as structured logs to an external buffer at each step of the inference process:

- Structural classification of inputs (coordinate assignment to intent, emotional, contextual layers, etc.)
- Recording of intermediate representations at each inference step (time-series tracking of structural position)

- Output provenance mapping (recording which part of the output corresponds to which input structure)
- Automatic detection of structural deviations (flags for departures from normal inference flow)

Logs are output in JSON-LD format and retained as provenance records compliant with the W3C PROV-O ontology in a state analyzable by third parties. The model body and log buffer operate in separate process spaces, making it architecturally impossible for the model to directly manipulate the logs.

4.3 Hardware Layer: Ensuring Physical Immutability

Software layer implementation cannot completely eliminate the risk of the entire execution environment being overwritten. The highest-standard EVA requires physical immutability at the hardware layer. The following implementation forms are under consideration:

- Burning verification logic into FPGA (Field-Programmable Gate Array): Physical implementation unmodifiable by software updates
- Embedding in ASIC (Application-Specific Integrated Circuit): Highest level of tamper resistance. Suitable for large-scale deployment
- TPM (Trusted Platform Module) integration: Cryptographic integrity guarantee for verification logs
- Secure Enclave utilization: Isolated execution environment independent even from the host OS

This hardware implementation is functionally analogous to HSMs (Hardware Security Modules) in financial systems or flight recorders on aircraft. It represents a conceptual inversion: the 'black box' in AI systems functions socially as a verification log rather than as model inference.

5. International Standardization of EVA: An Indispensable Condition for Social Implementation

5.1 Why International Standardization is Indispensable

As long as EVA remains a proprietary implementation by individual AI developers, its social impact will be limited. For EVA to genuinely function as 'infrastructure for proving AI safety,' it must be standardized as a unified technical specification across industries and internationally.

There are three reasons why international standardization is indispensable. First, interoperability: for third-party institutions to compare and evaluate verification logs generated by AI systems from different vendors, common formats and evaluation criteria are necessary. Second, regulatory effectiveness: the 'external verification' demanded by the EU AI Act and ISO/IEC 42001 can only be operationalized as concrete technical requirements once a standardized EVA specification exists. Third, transferability of trust: EVA's credibility must be based on certification by independent standards bodies, not on the reputation of the implementing company.

5.2 Alignment Between EVA and Major Standards Bodies

Body / Framework	EVA Alignment	Specific Proposal
ISO/IEC JTC 1/SC 42 (AI International Standards)	ISO/IEC 42001 complementary standard	New Work Item Proposal for 'Technical Specifications for External Verification of AI Systems.' Defines log format, granularity, and retention requirements
CEN-CENELEC JTC 21 (EU AI Standardization)	Technical basis for EU AI Act Articles 12 and 13	Proposal to incorporate EVA technical specifications into European Standard (EN) corresponding to the log-keeping obligations for high-risk AI
IEEE SA (AI Ethics / Transparency)	Alignment with IEEE P2863 etc.	Position EVA as the technical verification method for AI system transparency and ethical compliance
NIST AI RMF (US AI Risk Management)	Technical fulfillment of MEASURE function	Align EVA as the technical implementation of the continuous risk measurement sought by the MEASURE category
ITU-T AI Focus Group	Application to telecom and infrastructure AI	Propose EVA as external verification requirements for AI systems embedded in telecommunications infrastructure

5.3 EVA's Proposal Strategy in the Standardization Process

A realistic standardization pathway for EVA proceeds through the following staged process:

Stage	Action	Purpose / Expected Outcome
1	Establish public technical documentation	Publish EVA technical specifications as DOI-assigned preprints (Zenodo / SSRN). Establish a citable foundation for international standards proposals
2	Technical proposal to national standards bodies	Submit EVA as a New Work Item Proposal to ISO/IEC SC 42 through Japan's JISC / Information Technology Standards Commission of Japan (ITSCJ)
3	Collaboration with European standards bodies	Propose EVA technical specifications to CEN-CENELEC JTC 21. Establish direct alignment with the EU AI Act technical foundation
4	Build implementation track record	Conduct proof-of-concept (PoC) in collaboration with AI companies. Establish empirical basis for standards proposals
5	Transfer to independent foundation	Transfer governance of EVA standard to an independent foundation with multi-stakeholder composition. Secure the neutrality and sustainability of the standard

5.4 Social Impact of Standardization

If EVA is established as an international standard, the following societal changes can be expected:

- EVA conformance will be explicitly specified as a technical requirement in AI system procurement (government procurement, healthcare, finance, and infrastructure sectors)
- A third-party certification market for AI safety will form, giving rise to an ecosystem of independent verification institutions
- Submitting 'proof of safety' rather than 'assertions of safety' will become standard practice for developers
- EVA conformance may become the basis for mutual recognition of AI products in international trade

6. Human Natural Structure (HNS): As a Technical Method for Realizing EVA

6.1 The Question of Reference Baselines in EVA Implementation

For EVA to function, the verification mechanism must answer the question: what criteria are used to judge the appropriateness of reasoning? Merely recording structured logs is insufficient — an independent reference baseline is required that can determine whether the reasoning indicated by those logs is aligned with human intent, context, and values.

This reference baseline must satisfy the following conditions: culturally and linguistically neutral; applicable across domains; defined in machine-readable format; and independent of developer subjectivity. No such structural reference baseline satisfying these conditions has been established in current AI research.

6.2 HNS's Suitability as an EVA Reference Baseline

The Human Natural Structure (HNS) framework (Hara, 2026) is one technical method that can concretely satisfy the conditions required of the reference baseline that EVA demands.

HNS defines a 36-cell structural coordinate system formed as the product of six natural layers (Physical / Perceptual / Internal / Intentional / Relational / Societal) and six cognitive categories (Existence / Perception / Interpretation / Intention / Action / Interaction) describing human cognition, behavior, and social meaning. This coordinate system is defined in machine-readable JSON-LD / RDF format, directly enabling programmatic third-party verification.

In relation to EVA's three conditions, HNS provides the following:

- Verifiability: The HNS coordinate system is defined in JSON-LD format and can output each step of reasoning as machine-readable logs
- Transparency: HNS can be implemented as a sidecar independent of model inference, structurally preventing the model body from manipulating HNS coordinate assignments
- Physical Immutability: The HNS coordinate system, Non-Mapping Rule, and Bypass Prevention Logic can be embedded in FPGA / ASIC, realizing hardware-level tamper resistance

6.3 The Positioning of HNS within EVA

It is important to note that HNS is not the only method for realizing EVA. EVA is a technical framework, and other structural architectures can in principle be adopted as its reference baseline.

Human Natural Structure (HNS) is **one technical method for realizing External Verification Architecture (EVA)**.

HNS is already publicly available on GitHub (https://github.com/satoru-hara/03_NSW), with specifications, implementation reports, and technical documentation available as public information. This means that the transparency requirements in the international standardization discussion are already satisfied at the current stage. For consideration of HNS as a reference baseline for EVA, the reader is referred to the HNS Manuscript (Hara, 2026). The detailed technical specifications, design principles, and application cases of HNS are reserved for a separate paper; the primary focus of this paper is the proposal of EVA's conceptual framework.

7. Conclusion

This paper proposed the concept of External Verification Architecture (EVA), taking as its starting point the structural problem of self-assessment dependency in AI system safety assurance. EVA is a technical framework for external verification defined by three structural conditions — Verifiability, Transparency, and Physical Immutability — and is positioned as an architecture that newly fulfills the functional layer of 'continuous structural external verification at inference time' that existing XAI, AI auditing, and model card approaches are structurally unable to provide.

EVA's technical implementation spans two layers: the software layer (sidecar architecture, structured logging) and the hardware layer (FPGA / ASIC / TPM). It is the integration of these two layers that fully satisfies EVA's three conditions.

International standardization is indispensable for EVA to function as social infrastructure. Without integration into ISO/IEC JTC 1/SC 42, CEN-CENELEC JTC 21, and the EU AI Act technical specifications, EVA will remain confined to individual implementations. The standardization pathway presented in this paper — establishing public technical documentation, proposing to national standards bodies, collaborating with European standards organizations, and building an implementation track record — constitutes a concrete blueprint for developing EVA into a genuine international standard.

As AI systems become increasingly entrenched as social infrastructure, the transition from 'asserting safety' to 'proving safety' is an inevitable challenge — both technically and institutionally. EVA aims to contribute to international standardization discussions as the architectural foundation that makes this transition possible.

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